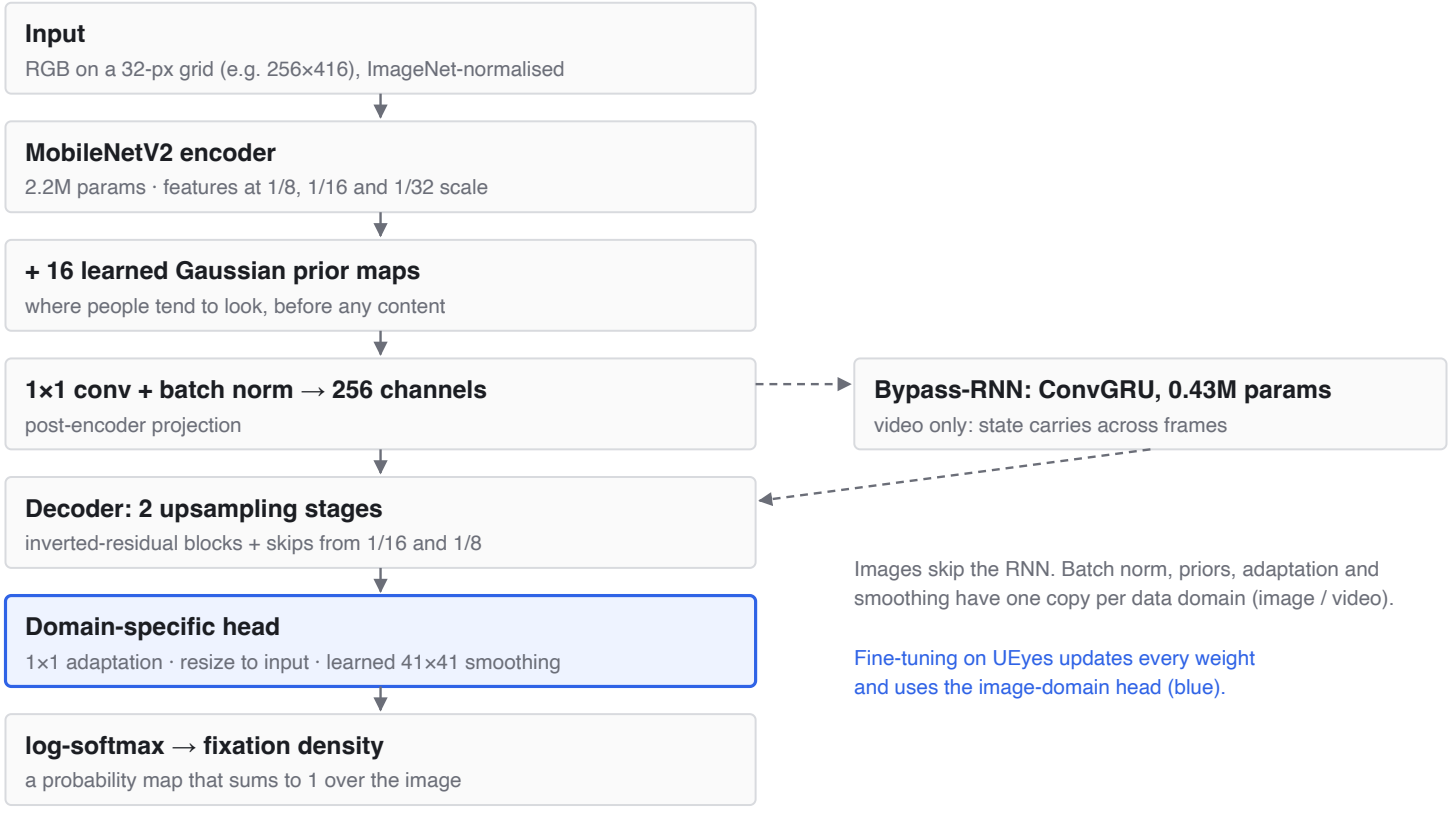
Input
RGB on a 32-px grid (e.g. 256×416), ImageNet-normalised
MobileNetV2 encoder
2.2M params · features at 1/8, 1/16 and 1/32 scale
+ 16 learned Gaussian prior maps
where people tend to look, before any content
1×1 conv + batch norm → 256 channels
post-encoder projection
Bypass-RNN: ConvGRU, 0.43M params
video only: state carries across frames
Decoder: 2 upsampling stages
inverted-residual blocks + skips from 1/16 and 1/8
Domain-specific head
1×1 adaptation · resize to input · learned 41×41 smoothing
log-softmax → fixation density
a probability map that sums to 1 over the image
Images skip the RNN. Batch norm, priors, adaptation and smoothing have one copy per data domain (image / video).
Fine-tuning on UEyes updates every weight and uses the image-domain head (blue).